

ACTIVITY - 4

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1) Given input {4371, 1323, 6173, 4199, 4344, 9679, 1989} and hash function $h(x) = x \pmod{10}$, Show the resulting:

- Separate chaining hash table
- open addressing hash table using linear probing
- open addressing hash table using quadratic probing
- open addressing hash table with second hash function $h_2(x) = 7 - (x \pmod{7})$.

$$h(4371) = 4371 \pmod{10} = 1$$

$$h(1323) = 1323 \pmod{10} = 3$$

$$h(6173) = 6173 \pmod{10} = 3$$

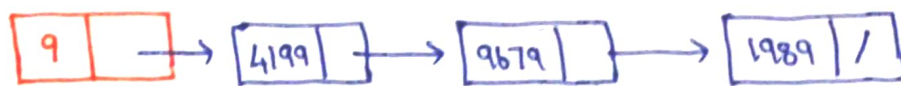
$$h(4199) = 4199 \pmod{10} = 9$$

$$h(4344) = 4344 \pmod{10} = 4$$

$$h(9679) = 9679 \pmod{10} = 9$$

$$h(1989) = 1989 \pmod{10} = 9$$

0	
1	→ [4371 /]
2	
3	→ [1323] → [6173 /]
4	→ [4344 /]
5	
6	
7	
8	
9	



Linear probing:

$$4371 \rightarrow 4371 \% 10 = 1$$

$$1323 \rightarrow 1323 \% 10 = 3$$

$$6173 \rightarrow 6173 \% 10 = 3 \rightarrow \text{Collision}$$

$$4199 \rightarrow 4199 \% 10 = 9$$

$$4344 \rightarrow 4344 \% 10 = 4$$

$$9679 \rightarrow 9679 \% 10 = 9 \rightarrow \text{Collision}$$

0	9679
1	4371
2	1989
3	1323
4	6173
5	4344
6	
7	
8	
9	4199

$$h(6173) \rightarrow (6173 + 1^2) \% 10 = 4 \times$$

$$(6173 + 2^2) \% 10 = 7$$

$$h(9679) \rightarrow (9679 + 1^2) \% 10 = 0$$

6173 will be store in index position 7 and
9679 will be store in index position 0

Quadratic probing:

$$h(4371) = 1$$

$$h(1323) = 3$$

$$h(6173) = 3$$

$$= (6173 + 1^2) \% 10 = 4$$

$$h(4199) = 9$$

$$h(4344) = 4$$

$$(4344 + 1^2) \% 10 = 5$$

$$h(9679) = 9$$

$$= (9679 + 1^2) \% 10 = 0$$

$$h(1989) = 9$$

0	9679
1	4371
2	
3	1323
4	6173
5	4344
6	
7	
8	1989
9	4199

$$= (1989 + 1^2) \% 10 = 0$$

$$= (1989 + 2^2) \% 10 = 3$$

$$= (1989 + 3^2) \% 10 = 8$$

Double hashing:-

Key	$h_1(x)$	$h_2(x)$
4371	1	4
1323	3	7
6173	3	1
4199	9	1
4344	4	3
9679	9	2
1989	9	6

$$h_2(4371) = 1$$

$$h_2(1323) = 3$$

$$h_2(6173) = 3$$

$$= 3 + 1(1) = 4$$

$$h_2(4199) = 9$$

$$h_2(4344) = 4$$

$$= 4 + 1(3) = 7$$

$$h_2(9679) = 9$$

$$= 9 + 1(2) = 1$$

$$= 9 + 2(2) = 8$$

$$= 9 + 3(2) = 5$$

$$h_2(1989) = 9 \Rightarrow 9 + 4(6) = 3, 5, 7, 1$$

Not inserted

0	
1	4371
2	
3	1323
4	6173
5	9679
6	
7	4344
8	
9	4199